Artificial Intelligence
Lecture 12: Natural Language Processing

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The same old song...

In real life, problems are formulated in natural language: English, French, ...

To be solved using AI techniques, it is essential to translate them into a mathematical formalism understandable by a machine.

Example: Mike is the son of Alice and Bob. He is Eve's brother. Who is Eve's mother?

We solved this problem by translating this sentence into propositional logic and using inference to answer the question.

Answering the question with propositional logic

Question: Mike is the son of Alice and Bob. He is Eve's brother. Who is Eve's mother?

Question in propositional logic:

```
son(mike, alice).
son(mike, bob).
brother(mike, eve).
mother(X, mike).
```

Knowledge base: Statements in propositional logic stating (at least) that siblings have the same parents, that a mother is a parent and a female and that Alice is a female.

Natural language processing

Natural language processing is the AI domain which consists in interacting directly at the level of natural language. This involves either (if not all) of the following:

- **Understanding:** The machine is able to interpret the sentence written by the user and to connect them to the existing knowledge base;
- **Producing:** The machine is able to generate texts in natural language that answer the queries of the user (and, preferably, are correct).

NLP nowadays...

Mike is the son of Alice and Bob. He is Eve's brother. Who is Eve's mother?

Given the information:

1. **Mike is the son of Alice and Bob.**
   - This means Alice is Mike's mother, and Bob is Mike's father.
2. **Mike is Eve's brother.**
   - This means Mike and Eve share at least one parent.

Since Alice is Mike's mother and Mike and Eve are siblings, they likely share the same mother.

Therefore, **Eve's mother is Alice.**

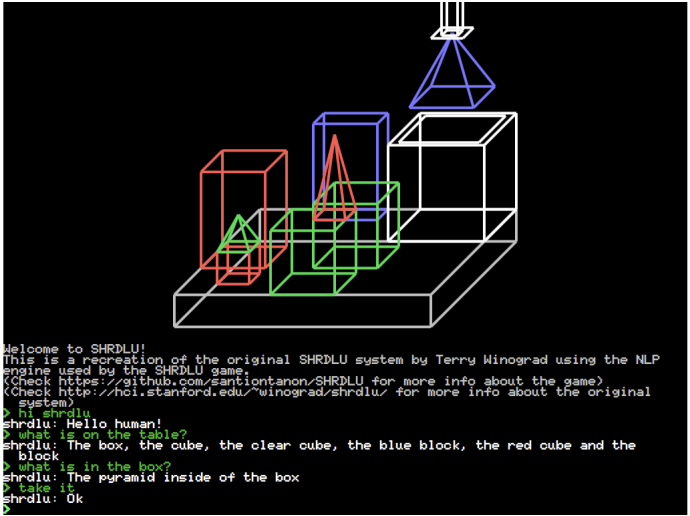
Eliza

```
  Welcome to
                    EEEEEE  LL      IIII  ZZZZZZZ   AAAAA
                    EE      LL       II        ZZ   AA   AA
                    EEEEE   LL       II      ZZZ    AAAAAAA
                    EE      LL       II     ZZ      AA   AA
                    EEEEEE  LLLLLL  IIII  ZZZZZZZ  AA   AA

  Eliza is a mock Rogerian psychotherapist.
  The original program was described by Joseph Weizenbaum in 1966.
  This implementation by Norbert Landsteiner 2005.


ELIZA: Please tell me what's been bothering you.
YOU:   Mike is the son of Alice and Bob. He is Eve's brother. Who is Eve's mothe
r?
ELIZA: Does that question interest you ?
YOU:   Yes, so please answer
ELIZA: I see.
YOU:   So... ?
ELIZA: Tell me more about that.
YOU:   I want to know who is Eve's mother
ELIZA: What does wanting to know who is eve's mother have to do with this discus
sion ?
YOU:   -_-
```

SHRDLU


Welcome to SHRDLU!
This is a recreation of the original SHRDLU system by Terry Winograd using the NLP engine used by the SHRDLU game.
(Check https://github.com/santiontanon/SHRDLU for more info about the game)
(Check http://hci.stanford.edu/~winograd/shrdlu/ for more info about the original system)
> hi shrdlu
shrdlu: Hello human!
> what is on the table?
shrdlu: The box, the cube, the clear cube, the blue block, the red cube and the block
> what is in the box?
shrdlu: The pyramid inside of the box
> take it
shrdlu: Ok
>

Back to our question

Question: Mike is the son of Alice and Bob. He is Eve's brother. Who is Eve's mother?

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It's normal: "Eve's" is derived from "Eve", it is the *Saxon genitive* form of it.

Identifying this is called **morphological analysis**.

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Identifying this is related to the problems of **part-of-speech tagging** and/or **parsing**.

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Problem 3: The word "He" is ambiguous: it can refer either to Mike or to Bob.

Identifying this is called **disambiguation** and is a **semantic** task.

Outline

1 Morphology

2 Part-of-speech tagging

3 Syntactic level: Grammars

Plan

1. Morphology
   - What is a word?
   - Morphology
   - Morphological parsing
2. Part-of-speech tagging
3. Syntactic level: Grammars

What is a word?

Intuitively, a **word** is the smallest element of language which can stand alone.

Examples: fish, car, university, be, was...

But...

- "cars" is not the shortest form: it can be reduced to a shorter element ("car")
- "Hauptstadt" is not the shortest form: it can be split into two shorter words: "haupt" and "Stadt".

We need to fine a better definition...

Orthographic definition

Another intuitive definition consists in observing that words are delimited by spaces and punctuation signs.

But...

- In Mandarin Chinese, characters are not separated.
- Vietnamese splits all syllables: For instance "sinh viên" (student)
- In English, some compound words may contain spaces: "ice cream", "get up"...
- In German, some words may be in two parts that are separated in the sentence: "ich *sehe* dich *an*"

Phonological definition

Since the written form does not help much, maybe the spoken form will...

A phonological vision defines a word as a unit larger or equal to one syllable and which can be distinguished based on phonological features (e.g. prosody, accents etc)

Example: In several language, the stress is always on a specific syllable: the first one (Begali, Czech, Estonian, Icelandic, Kongo, Saami...), the last one (Khmer, Persian, Tajik...) or other (e.g. 3rd one for Winnebago)

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But...

- In English, "I'll" reads in one syllable but corresponds to two words.
- In Finnish, "pääkaupunki" is one word but contains two distinct phonological words (based on the phonological rule of vowel harmony)

Back to the question: What is a word?

There is no consensus on a definition of what a word is in linguistic!

Any attempt finds some counter-example.

Remark: A general risk in linguistic is to adopt a Western-centric vision, and to forget about the diversity of languages.

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- What is a word?
- Morphology
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Morphemes

Morphology is the field of linguistic that studies the internal structure of words.

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These words are all built upon a base word (“speak”), called **stem**, to which are added some elements (prefixes and suffixes).

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A **morpheme** is a minimal unit of meaning or grammatical function in a word.

Example: The word "reopens" decomposes as "re-" + "open" + "-s".

- "re-" is a prefix and corresponds to a unit of meaning.
- "open" is the stem and corresponds to a unit of meaning.
- "-s" is a unit of grammatical function.

An example: Morphology in Finnish

Käynnistyttelijä
He who makes something start up itself repeatedly
Käynnistely
Repetitive starting up
Käynnistytys
The act of making something start up itself
Käynnistytellä
To make something start up itself repeatedly
Käynnistelijä
He who starts up (repeatedly)
Käynnistyttelyt
The act of making something start up itself repeatedly
Käynnistyttää
To make something start up itself
Käynnistyttäjä
He who makes something start up itself
Käynnistyä
To start up itself
Kävelyttää
To walk (transitive)
Käynnistellä
To start up (repeatedly)
Käynnistäjä
He who starts up
WELCOME TO THE
FINNISH LANGUAGE
Kävely
Walk
Kävelijä
Walker
Käynnistää
To start up
Käynnistys
Starting-up
Kävellä
To walk (intransitive)
Käyttely
Repetitive use
Käyttelijä
He who uses repetitively
Käyttäytyminen
Behavior
-käyttöisyys
Untranslatable
Käyttäytyä
To behave
Käytellä
To use (repeatedly)
Käyttönällistely
The act of making practical (repeatedly)
Käyttönällistelijä
He who makes practical (repeatedly)
Käyttönällistelemällä
With the one that was made practical
Käyttönällistelemällä
With the one that he made practical
Käyttönällistelemälläänkin
With the one that he made practical too
Käyttönällistelemälläänkinkö
With the one that he made practical too?
Käyttönällistelemälläänkinköhän
With the one that he made practical too, I wonder?
Käyttönällistä
The act of making practical
Käyttönällistäjä
He who makes practical
Käyttönällistä
To make practical
Käyttönällisyys
Practicality
Käyttönällisin
Practical
Käyttönällisimmin
The most practically
Käyttönällisemmin
More practically
Käyttönällisesti
Practically
Käyttönällisyys
Practicality
Käyttönällisyyttä
Practicality
Käyttönällisyyttä
Practicality
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Derivational and inflectional morphology

Derivational morphemes are used to make new words from the stem:

Example: good $\rightarrow$ goodness, conventional $\rightarrow$ unconventionally, mind $\rightarrow$ mindfulness...

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Example: good $\rightarrow$ goodness, conventional $\rightarrow$ unconventionally, mind $\rightarrow$ mindfulness...

Inflectional morphology is instead used to indicate aspects of the grammatical function of a word (tense, case, plural, possessive, comparative...)

Example: dog $\rightarrow$ dogs, play $\rightarrow$ played, bright $\rightarrow$ brightest...

Morphology in NLP

Using morphology is an important task in Natural Language Processing:

- Morphological transformations might increase the sparsity: some morphological transformations (inflectional or derivational) can appear only rarely, if not once.
  - *Inflectional:* In Finnish, koirallenikinkohan = "to my dog too, right?"
  - *Derivational:* In English, neologisms like cheesecakism, bighousedness...
- Language generation: Knowing that I want to use the verb "play", present sg 3rd person, which form should I use?

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Morphological parsing

The task of morphological parsing is the following:

Input: a word

Output: the word's stem + features expressed by other morphemes

Examples:

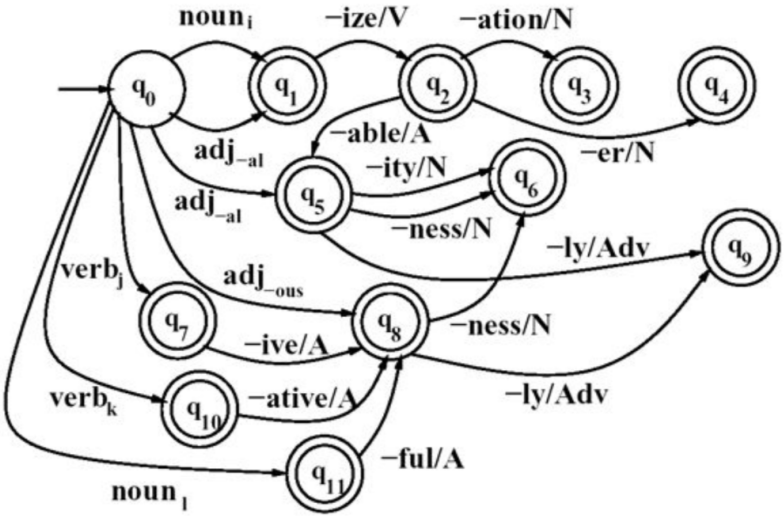
- universities $\rightarrow$ {university +N +Pl }
- dogs $\rightarrow$ {dog +N +Pl, dog +V +3P +Sg }

Finite-state automata

A **Finite-state automaton** encodes a set of strings that can be obtained by following paths from a start state to a final state in the following graph:

- Q: finite set of states
- q_0: start state
- $F \subseteq Q$: set of final states
- Σ: finite alphabet
- Transitions: $Q \times \sigma^* \rightarrow Q$

FSA for a fragment English morphology



From representing words to representing morphological transformations

FSA is designed to represent a set of strings, e.g. { walk, walks, like, liked }.

In morphology, we are interested in representing pairs of strings, e.g. (walk+V+3P+Sg, walks) or (geese+N+Sg, goose).

From representing words to representing morphological transformations

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In morphology, we are interested in representing pairs of strings, e.g. (walk+V+3P+Sg, walks) or (geese+N+Sg, goose).

In such pairs, the two words could be represented independently, one after the other, but we would lose some information about common structure.

Example: (cat+N+Pl, cats) has the common string "cat" shared between the input and output:

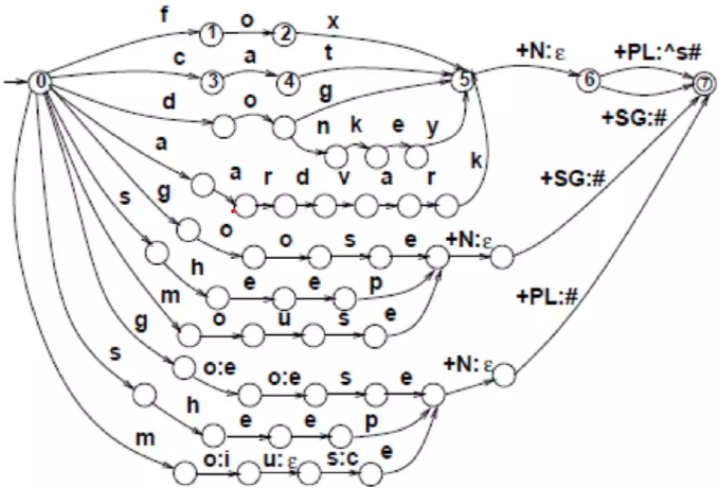
c a t +N:s +Pl:ε

Finite-state transducers

A **Finite-state automaton** encodes a set of strings that can be obtained by following paths from a start state to a final state in the following graph:

- Q: finite set of states
- q_0: start state
- $F \subseteq Q$: set of final states
- Σ: finite alphabet (input)
- Δ: finite alphabet (output)
- Transition function: $Q \times \sigma^* \rightarrow Q$
- Output function: $Q \times \sigma^* \rightarrow \Delta^*$

Example: FST for plural of nouns



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An illustrative example

They can fish

An illustrative example

They can fish

Meaning 1: They know how to fish. In this case:

- they = pronoun
- can = modal verb
- fish = verb

An illustrative example

They can fish

Meaning 1: They know how to fish. In this case:

- they = pronoun
- can = modal verb
- fish = verb

Meaning 2: They put fish in cans. In this case:

- they = pronoun
- can = verb
- fish = noun (plural)

Plan

1 Morphology

2 Part-of-speech tagging
- Defining a tag set
- Statistical POS tagging

3 Syntactic level: Grammars

Defining a tag set

A **tag** corresponds to a label for a word class. A tag set is an inventory for such tags.

The granularity of the tagset depends on the problem and of the pursued goal.

Several standard tagsets exist, usually introduced in some published datasets:

- Brown tagset (87 tags)
- Penn Treebank (45 tags, standard)
- CLAWS 5 (62 tags)

Remark: POS tagsets depend on the language: For instance FinnTreeBank2 for Finnish (with... 612 tags!)

Word classes (open)

- **Nouns:**
  - NN: singular (*dog*) or mass (*snow*)
  - NNS: plural (*dogs*)
  - NNP: singular proper noun (*Germany*)
  - NNPS: plural proper noun (*Smiths*)
- **Verbs** (in different morphological forms):
  - VB: infinitive form (*walk*)
  - VBD: past tense (*walked*)
  - VBZ: 3rd person singular present tense (*walks*)
  - ...
- **Adjectives:**
  - JJ: base form (*big*)
  - JJR: comparative (*bigger*)
  - JJS: superlative (*biggest*)
- **Adverbs:** RB (*slowly*) / RBR (*slower*) / RBS (*slowest*)

Word classes (close)

- **Modal verbs:** MD, copula (*be*), modal (*can*, *may*, *might*...) and auxiliaries (*be*, *have*, *will*, *do*)
- **Preopositions:**
  - IN: preposition (*on*, *with*, *by*, *despite*...)
  - TO (*to* before a verb)
- **Conjunctions:**
  - CC: coordination (*and*, *or*, *but*)
  - IN: subordination (*that*, *whether*...)
- **Particles:** RP (*on*, *over*, *up*... depending on a verb)
- **Pronouns:**
  - PRP: Personal (*I*, *you*, ...)
  - PRP$: Possessive (*my*, *your*, ...)
  - WP: wh-pronoun (*who*, *which*, ...)
- **Determiners:** DT articles (*the*, *an*, *a*), demonstratives (*this*...), quantifiers (*some*, *every*...)

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POS tagging as a statistical problem

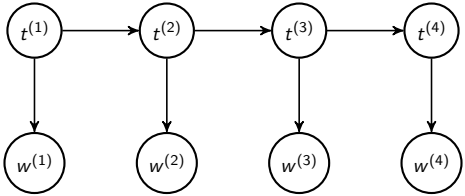
$$
\begin{array}{rcccccc}
 & \text{She} & \text{promised} & \text{to} & \text{back} & \text{the} & \text{bill} \\
\mathbf{w} = & w^{(1)} & w^{(2)} & w^{(3)} & w^{(4)} & w^{(5)} & w^{(6)} \\
\mathbf{t} = & t^{(1)} & t^{(2)} & t^{(3)} & t^{(4)} & t^{(5)} & t^{(6)}
\end{array}
$$

The goal is to determine the tag sequence $\mathbf{t}$ that is the most likely to explain the observed word sequence $\mathbf{w}$.

Using probabilistic language:

$$\mathbf{t}^* = \arg\max_{\mathbf{t}} P(\mathbf{t}|\mathbf{w})$$

Defining the probabilistic model: Hidden Markov Model



$$P(\mathbf{t}|\mathbf{w}) = P(\mathbf{t})P(\mathbf{w}|\mathbf{t})$$

with:

- the prior over $\mathbf{t}$ given as a bigram: $P(\mathbf{t}) = \prod_i P(t^{(i+1)}|t^{(i)})$
- the likelihood $P(\mathbf{w}|\mathbf{t}) = \prod_i P(w^{(i)}|t^{(i)})$

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Until now...

We have seen:

- how to represent and encode the **morphological transformations** of words;
- how to identify the functions (parts of speech) of words in a text.

We will now investigate how to represent the syntax of sentences, i.e. which sentences are syntactically correct.

Plan

1 Morphology

2 Part-of-speech tagging

3 Syntactic level: Grammars
- Representing the sentence structure
- Context-free grammars
- Beyond CFGs

A simple sentence

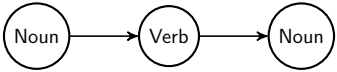
I eat pizza

This sentence can be decomposed as:

- I: noun (subject)
- eat: verb
- pizza: noun (object)

Sentence structure as a finite-state automaton

Based on this example, we can propose a first simple grammar using a finite state automaton:



FSAs can be used to generate various other structures.

Example: Write a FSA describing the fact that a noun can be preceded by a determiner and as many adjectives as we want.

The problem with FSAs: Attachement ambiguity

I am watching a movie with Clint

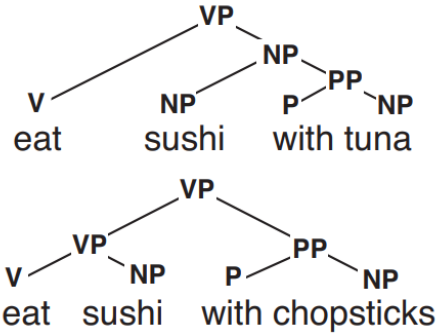
What is the structure of this sentence? Here it is ambiguous: to what does "with Clint" connect?

- **Structure 1:** I am watching (a movie) (with Clint). $\Rightarrow$ in the company of Clint
- **Structure 2:** I am watching (a movie, with Clint). $\Rightarrow$ a movie in which Clint is starring

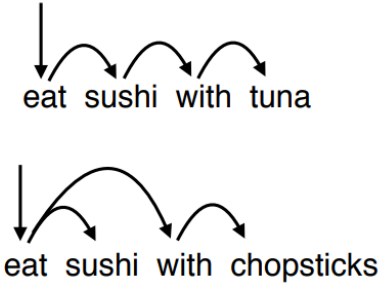
FSA are not sufficient to describe the grammatical construction of the sentence since they do not allow to explicit the dependencies between words / constituents of the sentence.

Representing phrase structures

Phrase structure trees



Dependency trees



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Context-free grammar

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- N is a set of **non-terminals**, representing the different type of phrases or clauses in the language, e.g. $N = \{S, N, V, NP, VP, \dots\}$

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- R is a set of rules of the form $A \rightarrow \beta$, where $A \in N$ and $\beta \in (N \cup \Sigma)^*$, e.g.

$$N \rightarrow \{\text{university, garden, sushi}\}$$
$$DT \rightarrow \{\text{a, the}\}$$
$$NP \rightarrow DT\ N$$

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$$\begin{aligned} N &\rightarrow \{\text{university, garden, sushi}\} \\ DT &\rightarrow \{\text{a, the}\} \\ NP &\rightarrow DT\ N \end{aligned}$$

- $S \in N$ is the **start symbol**, designating the whole sentence.

Different kinds of phrases

There are different kinds of constituents:

- **Noun phrases:** the man, a dog, Germany...

Different kinds of phrases

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- **Prepositional phrases:** with glasses, in the university...

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- **Noun phrases:** the man, a dog, Germany...
- **Prepositional phrases:** with glasses, in the university...
- **Verb phrases:** eat pizza, drink, learn quickly...

Representing noun phrases

Let us start by listing a few noun phrases in English:

[He] drinks

[Bob] drinks

[A student] drinks

[A bad student] drinks

[The student in the back] drinks

[The student who hates studies] drinks

Representing noun phrases

This translates into the following rules:

He drinks	(pronoun)	NP $\rightarrow$ Pronoun
Bob drinks	(proper noun)	NP $\rightarrow$ ProperNoun
A student drinks	(det + noun)	NP $\rightarrow$ Det Noun
A bad student drinks	(det + adj + noun)	NP $\rightarrow$ AdjP Noun
The student in the back drinks	(NP + PP)	NP $\rightarrow$ NP PP
The student who hates studies drinks	(NP + relative clause)	NP $\rightarrow$ NP RelClause

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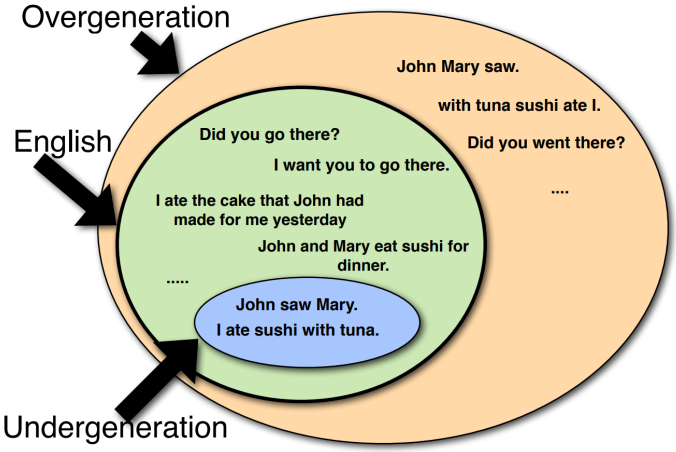
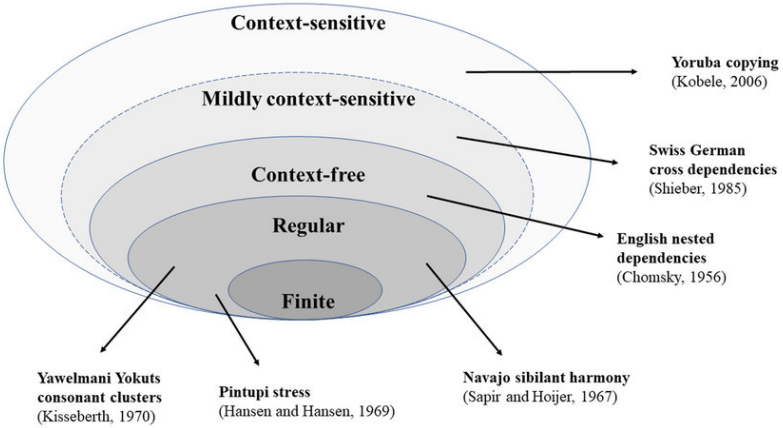
Grammars


Overgeneration
English
Undergeneration
John Mary saw.
with tuna sushi ate I.
Did you went there?
....
Did you go there?
I want you to go there.
I ate the cake that John had made for me yesterday
John and Mary eat sushi for dinner.
.....
John saw Mary.
I ate sushi with tuna.

Why go beyong CFGs?

CFGs capture only **nested** dependencies: The dependencies do not cross.

Example: "... that I saw (Anna let (Mary help (Hans swim)))

- **In German:** ... dass ich (Anna (Maria (Hans schwimmen) helfen) lassen) sah
- **In Dutch:** ... dat ik Anna Maria Hans zag laten helpen zwemmen

Chomsky hierarchy